The Long Run: Scope of the Main Paper

Companion note to the manuscript

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1 The scope

The paper is a Nash-equilibrium analysis. Every equilibrium in it is an equilibrium of a game played once: the incumbent's identity is fixed, the prize is awarded, and the model stops (manuscript, Appendix D, "The contest"). Over time the contest repeats. The winner holds the position next period, the displaced holder competes again, and wealth responds to past success.

The two are read together as forces and states. At any date the state is the wealth profile, who holds the position, and the prize, the fee and the weight in force. The one-shot equilibrium at that state is where the opposing forces of the model settle. The fee weighs less on the rich, so they enter first (K1, K3); the gain from entering falls as more enter, which caps entry (M4, K4); a spread of wealth moves the count in a direction set by where the marginal entrant stands (K2); a heavier weight on the base component lowers entry (K8) without guaranteeing a better winner (M32). The long run decides which states are reached. Its variables are not free parameters but values the process arrives at, and every claim of the paper holds at whatever values it reaches. The claims are therefore conditional on the state, which is the scope this note records.

The bridge condition. The reading holds when the period game is the one-shot game. That is so if agents are myopic, or if the value of holding the position after this period is the same for every player. A forward-looking challenger values winning at V plus the discounted value of incumbency. Since V enters the gain only as a scale (M1), a continuation value common to all players changes the prize and nothing else, and every result holds with the larger prize. If the continuation value depends on a player's own wealth, the prize is no longer valued alike by rich and poor (Appendix D, "Payoffs"), and wealth acts through the prize as well as through the fee. The period game is then a different game, and that is the one case in which the long run would change the claims themselves rather than the state at which they are read.

2 What the one-shot model fixes

Given the wealth profile and the incumbent at a date, the one-shot results hold at that date. The table lists the ones the extension would lean on.

Statement	Content	Rests on
The count each period	Given the profile, every one-shot equilibrium has k^* entrants, and the k^* richest entering is one of them.	K1, M6, M8
Wealth enters only through κ	The fee's utility cost $\kappa(w) = u(w) - u(w-c)$ falls with wealth.	K3, M23
The prize is outside u	The prize enters payoffs linearly and outside the utility of wealth, so it does not change wealth.	Appendix D, Payoffs
The incumbent's entry	As player 0 with the lowest cost, she is in every non-empty equilibrium outside the band of M6.	M30
The entrants' share	The prize goes to an entrant with probability $\xi(k)$, non-decreasing in k, equal to 1 when all enter, non-increasing in μ.	M31
A common bonus	It washes out when every player holds it and favours holders once some stop buying.	M32

3 The ingredients of a steady state

Three choices turn the one-shot game into a process. Each is a modelling fork, and the extension must take one branch of each.

Turnover. The winner at date t is the incumbent at $t+1$; the displaced holder joins the challengers. Variants: the position lasts a fixed number of periods, or a replacement shock ends tenure.

The law of motion of wealth. In the manuscript the prize is outside u, so winning changes nothing but the payoff. For wealth to respond to success the position must pay. Two branches.

- **S-A, the position pays.** The holder receives an income $y > 0$ each period of tenure, in wealth units, besides the prize's value V in utility. Then $w_{t+1} = w_t - c\,e_t + y\,\mathbf{1}\{\text{holds}\}$, plus any endowment shock. This is a step from the privilege contest of Appendix D towards a rent-seeking contest, since part of what is won is now money.
- **S-B, the position does not pay.** Wealth moves only through the fees paid and an exogenous endowment process. Then every entrant is poorer next period by c, and success does not feed wealth.

The life of the components. Either the base and bought components are redrawn each period (memoryless), or the bought component persists for the holder once bought (a credential kept). Under persistence the fee is paid once, which is an investment rather than a fee, and the incumbent is scored as an entrant by inheritance rather than by choice.

4 How the claims change

The one-shot claims survive as statements about each period, since each period is the one-shot game at the current state. What changes is the state. The table gives, for each headline, what is unchanged and what the steady state adds, with the conjectures labelled.

Table 1: Each headline, per period and in the steady state.

Row	Per period	In the steady state
K1	Unchanged: the count is pinned at each date by the current profile.	The incumbent's identity becomes a Markov chain on the players. **S-1** (one step from M31 and the symmetry of the entrant law): with components redrawn each period and the incumbent scored as an entrant, the $k+1$ entrants win with equal probability, so the incumbent is displaced with probability at least $1-\xi(k)/(k+1)$, which is at least $\frac{1}{2}$ whenever one challenger enters, and with probability $1-\xi(0)$ when none does. Closure under redraw therefore needs $k^*=0$ and $\xi(0)=1$, which holds when $(1-\mu)s$ exceeds μ times the width of the support of r almost surely, as at the $\mu=0$ boundary with $s>0$. *False if* the entrant law is not exchangeable across entrants, which the anonymity of M26 rules out.
K2	Unchanged for a change of the current profile.	The spread is no longer exogenous. Under S-A the holder's wealth rises by y each period, so the stationary profile is more dispersed than the endowment profile. **S-2** (conjecture): the stationary dispersion of wealth rises in y. *False if* turnover is so fast that no holder accumulates, which S-1 makes possible at small $\xi(k)$. The sign rule of K2 then applies to a spread the process generates, read at the stationary marginal entrant.
K3	Unchanged.	Unchanged; κ is still the only channel within a period. Under S-A wealth also records past success, so the richest-first sorting of K1 reads as a sorting by past success.
K4, K5	Unchanged.	The prize gains a second channel. **S-3** (conjecture, S-A): the stationary count is non-decreasing in y, because a displaced holder returns richer by her tenure's income, and by M5 a richer challenger enters whenever a poorer one does. *False if* the income y raises the holder's wealth enough to change the law of s or r, which the anonymity of M26 excludes, or if y is taxed into the fee.
K6	Unchanged.	Unchanged; the number of challengers is fixed.
K7, K8	Unchanged.	**S-4** (conjecture, persistence): with the bought component kept by the holder, a higher μ lowers the number who buy (K8), and the holder's kept bonus then counts against fewer holders, so tenure lengthens and mobility falls; closure can then arise at $\mu>0$. Under redraw there is no such persistence, and S-1 bounds mobility below. *False if* the kept bonus depreciates faster than the entry response.
M30	The incumbent chooses as player 0.	**S-5** (one step from M5 and the law of motion): under S-A each period of tenure adds y to the holder's wealth while no challenger's wealth rises, so after enough periods of tenure the holder is the richest player, has the lowest cost and, by M5, enters whenever any challenger does. "Scored as an entrant" is then the outcome of the process for a holder of long enough tenure, not an assumption. *False if* tenure ends before that, which S-1 makes likely under redraw whenever a challenger enters.
M32	Unchanged.	The common-bonus mechanism compounds. Under persistence, a holder who keeps the bonus wins against outsiders with better base components in every period in which some challengers have stopped buying, so the winner's base component in the steady state is below its one-shot level. **S-6** (conjecture): under persistence and S-A, the stationary expected base component of the holder falls in μ. *False if* S-1's turnover dominates, as under redraw.
C1 to C14	Unchanged; they are claims about the one-shot model.	Unchanged as claims about each period; any claim about the steady state itself would be a new claim.

5 Mobility and closure as objects

Mobility is the stationary probability that the position changes hands. Closure is mobility zero. In the one-shot model closure is the $\mu = 0$ boundary, where outsiders score nothing (manuscript, Research question). In the process it has two further routes, each a conjecture above: a holder's wealth that makes her the lowest-cost player every period (S-5, which gives persistence of entry, not of the position), and a kept bonus that counts against fewer holders as μ rises (S-4, which gives persistence of the position). S-1 says that under redraw neither route closes the position while any challenger enters. So the sharpest statement the extension can aim at is that closure needs either the $\mu = 0$ boundary or a bought component that persists; wealth alone, under redraw, does not close it.

6 What this gives the manuscript

Two sentences in Appendix D of the manuscript ("The contest"). The first says that "equilibrium" means the one-shot equilibrium with the incumbent's identity fixed, so the claims are conditional on the state, and that a steady state is outside the model. The second gives the bridge condition, myopia or a continuation value common to all players, under which each period of a repeated contest is the one-shot game.

References